

Question 1: Linear Regression with Mean Absolute Error (MAE)

**Guess and Check** (inspect plotted points to find a candidate line achieving  $\mathcal{L}_{\text{MAE}} < 1.0$ ).  
Given the 4 training samples below, fit a linear model  $\hat{y} = wx + b$  that minimizes the Mean Absolute Error loss:

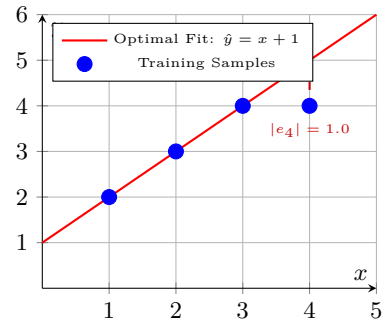
$$\mathcal{L}_{\text{MAE}}(w, b) = \frac{1}{N} \sum_{i=1}^N |(wx_i + b) - y_i|$$

(Note: If multiple optimal lines achieve the same minimum MAE loss, select the line that interpolates the maximum number of training points.)

Training Dataset:

| Sample ( $i$ ) | Feature ( $x_i$ ) | Target ( $y_i$ ) |
|----------------|-------------------|------------------|
| 1              | 1.0               | 2.0              |
| 2              | 2.0               | 3.0              |
| 3              | 3.0               | 4.0              |
| 4              | 4.0               | 4.0              |

Plot data points and sketch the optimal line:



Tasks & Questions:

a) Find the optimal parameters  $(w^*, b^*)$  that minimize MAE.

$w^* = \underline{1.0} \quad b^* = \underline{1.0}$

b) Calculate the minimum total MAE loss ( $\mathcal{L}_{\text{MAE}}^*$ ).

$\mathcal{L}_{\text{MAE}}^* = \underline{0.25}$

c) Compute predictions  $\hat{y}$  for the test points:

- $x = 1.5 \implies \hat{y} = \underline{2.5}$
- $x = 2.5 \implies \hat{y} = \underline{3.5}$
- $x = 3.5 \implies \hat{y} = \underline{4.5}$

Detailed Step-by-Step Solutions & Explanations:

• **Finding the Best Line by Guess and Check:** For Mean Absolute Error (MAE), we find the best-fitting line by **guess and check** (inspecting the plotted training data points to formulate a candidate model, then checking its residuals):

- Notice that the first three samples  $(1, 2), (2, 3), (3, 4)$  are collinear with slope  $w = \frac{3-2}{2-1} = 1.0$  and intercept  $b = 2.0 - 1.0(1.0) = 1.0$ .
- We guess the candidate model:  $\hat{y} = 1.0x + 1.0 \implies \mathbf{w^* = 1.0, \quad b^* = 1.0}$ .

• **Check Residuals and Compute Minimum MAE Loss ( $\mathcal{L}_{\text{MAE}}^*$ ):** Evaluating the absolute error  $|e_i| = |\hat{y}_i - y_i|$  for each training sample:

$$e_1 = |2.0 - 2.0| = 0.0, \quad e_2 = |3.0 - 3.0| = 0.0, \quad e_3 = |4.0 - 4.0| = 0.0, \quad e_4 = |5.0 - 4.0| = 1.0$$

Calculate the total MAE loss across the  $N = 4$  samples:

$$\mathcal{L}_{\text{MAE}}^* = \frac{1}{N} \sum_{i=1}^N |e_i| = \frac{0.0 + 0.0 + 0.0 + 1.0}{4} = \frac{1.0}{4} = \mathbf{0.25}$$

This candidate achieves the minimum possible loss while interpolating the maximum number of points (3 out of 4 samples).

• **Predictions for Test Points:** Using the fitted line  $\hat{y} = 1.0x + 1.0$ :

$$x = 1.5 \implies \hat{y} = 1.0(1.5) + 1.0 = \mathbf{2.5}, \quad x = 2.5 \implies \hat{y} = 1.0(2.5) + 1.0 = \mathbf{3.5}, \quad x = 3.5 \implies \hat{y} = 1.0(3.5) + 1.0 = \mathbf{4.5}$$

Question 2: Linear Regression with Mean Squared Error (MSE)

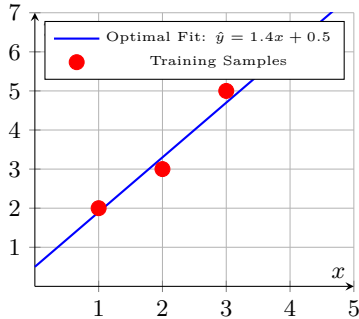
Closed-Form Solution / Normal Equations (derive optimal  $w^*, b^*$  analytically using sample means  $\bar{x}, \bar{y}$ ).  
Given the 4 training samples below, fit a linear model  $\hat{y} = wx + b$  that minimizes the Mean Squared Error loss:

$$\mathcal{L}_{\text{MSE}}(w, b) = \frac{1}{2N} \sum_{i=1}^N ((wx_i + b) - y_i)^2$$

Training Dataset:

| Sample ( $i$ ) | Feature ( $x_i$ ) | Target ( $y_i$ ) |
|----------------|-------------------|------------------|
| 1              | 1.0               | 2.0              |
| 2              | 2.0               | 3.0              |
| 3              | 3.0               | 5.0              |
| 4              | 4.0               | 6.0              |

Plot data points and sketch the optimal line:



Tasks & Questions:

a) Derive the optimal weights ( $w^*, b^*$ ) using analytical formulas ( $\bar{x}, \bar{y}$ ).

$w^* = \underline{\hspace{1cm} 1.4 \hspace{1cm}} \quad b^* = \underline{\hspace{1cm} 0.5 \hspace{1cm}}$

b) Calculate the minimum total MSE loss ( $\mathcal{L}_{\text{MSE}}^*$ ).

$\mathcal{L}_{\text{MSE}}^* = \underline{\hspace{1cm} 0.025 \hspace{1cm}}$

c) Compute predictions  $\hat{y}$  for the test points:

- $x = 1.5 \implies \hat{y} = \underline{\hspace{1cm} 2.6 \hspace{1cm}}$
- $x = 2.5 \implies \hat{y} = \underline{\hspace{1cm} 4.0 \hspace{1cm}}$
- $x = 3.5 \implies \hat{y} = \underline{\hspace{1cm} 5.4 \hspace{1cm}}$

Detailed Step-by-Step Solutions & Explanations:

- Closed-Form Ordinary Least Squares (OLS) Formulation:** Unlike MAE in Problem 1 (which required inspecting and guessing candidate lines due to non-smooth absolute values), the MSE loss  $\mathcal{L}_{\text{MSE}}(w, b) = \frac{1}{2N} \sum_{i=1}^N ((wx_i + b) - y_i)^2$  is smooth and strictly convex. Setting partial derivatives to zero ( $\frac{\partial \mathcal{L}}{\partial w} = 0, \frac{\partial \mathcal{L}}{\partial b} = 0$ ) yields exact **closed-form analytical normal equations**:

$$w^* = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}, \quad b^* = \bar{y} - w^* \bar{x}$$

- Analytical Derivations of Means and Variance/Covariance:**

- Sample Means:**  $\bar{x} = \frac{1.0 + 2.0 + 3.0 + 4.0}{4} = \mathbf{2.5}, \quad \bar{y} = \frac{2.0 + 3.0 + 5.0 + 6.0}{4} = \mathbf{4.0}$
- Computation Table for Centered Terms:**

| Sample ( $i$ )   | $x_i$ | $y_i$ | $(x_i - \bar{x})$ | $(y_i - \bar{y})$ | $(x_i - \bar{x})^2$     | $(x_i - \bar{x})(y_i - \bar{y})$ |
|------------------|-------|-------|-------------------|-------------------|-------------------------|----------------------------------|
| 1                | 1.0   | 2.0   | -1.5              | -2.0              | 2.25                    | +3.00                            |
| 2                | 2.0   | 3.0   | -0.5              | -1.0              | 0.25                    | +0.50                            |
| 3                | 3.0   | 5.0   | +0.5              | +1.0              | 0.25                    | +0.50                            |
| 4                | 4.0   | 6.0   | +1.5              | +2.0              | 2.25                    | +3.00                            |
| Sum ( $\Sigma$ ) | 10.0  | 16.0  | 0                 | 0                 | $S_{xx} = \mathbf{5.0}$ | $S_{xy} = \mathbf{7.0}$          |

3. Optimal Parameters:

$w^* = \frac{S_{xy}}{S_{xx}} = \frac{7.0}{5.0} = \mathbf{1.4}, \quad b^* = \bar{y} - w^* \bar{x} = 4.0 - (1.4)(2.5) = 4.0 - 3.5 = \mathbf{0.5}$

Fitted regression model:  $\hat{\mathbf{y}} = \mathbf{1.4x + 0.5}$ .

- Minimum Total MSE Loss Calculation:**

$i = 1 : \hat{y}_1 = 1.4(1.0) + 0.5 = 1.9 \implies e_1 = -0.1 \implies e_1^2 = 0.01 \quad i = 2 : \hat{y}_2 = 1.4(2.0) + 0.5 = 3.3 \implies e_2 = +0.3 \implies e_2^2 = 0.09$   
 $i = 3 : \hat{y}_3 = 1.4(3.0) + 0.5 = 4.7 \implies e_3 = -0.3 \implies e_3^2 = 0.09 \quad i = 4 : \hat{y}_4 = 1.4(4.0) + 0.5 = 6.1 \implies e_4 = +0.1 \implies e_4^2 = 0.01$

Sum of squared errors:  $\sum_{i=1}^4 e_i^2 = 0.01 + 0.09 + 0.09 + 0.01 = \mathbf{0.20}$ .  
Using the specified loss formula with  $\frac{1}{2N}$  scaling ( $N = 4$ ):

$$\mathcal{L}_{\text{MSE}}^* = \frac{1}{2 \times 4} \sum_{i=1}^4 e_i^2 = \frac{0.20}{8} = \mathbf{0.025}$$

- Test Point Predictions:**

- $x = 1.5 \implies \hat{y} = 1.4(1.5) + 0.5 = 2.1 + 0.5 = \mathbf{2.6}$
- $x = 2.5 \implies \hat{y} = 1.4(2.5) + 0.5 = 3.5 + 0.5 = \mathbf{4.0}$
- $x = 3.5 \implies \hat{y} = 1.4(3.5) + 0.5 = 4.9 + 0.5 = \mathbf{5.4}$

Question 3: Binary Logistic Regression

**Decision Boundary & Sigmoid Probability Inference** (evaluate  $z = 0$  half-spaces,  $\hat{p} = \sigma(z)$ , and BCE loss). Consider a binary classification task with 2D feature vector  $\mathbf{x} = [x_1, x_2]^T$ , logit equation  $z = w_0 + w_1x_1 + w_2x_2$ , and predicted probability  $\hat{p} = \sigma(z) = \frac{1}{1+e^{-z}}$ . The decision boundary is defined by  $x_1 + x_2 = 5$  with optimal parameters  $\mathbf{w}^* = [w_0^*, w_1^*, w_2^*]^T = [5.0, -1.0, -1.0]^T$ .

Training Dataset:

| Sample ( $i$ ) | $x_1$ | $x_2$ | Class ( $y_i$ ) |
|----------------|-------|-------|-----------------|
| 1              | 1.0   | 2.0   | 1               |
| 2              | 2.0   | 1.0   | 1               |
| 3              | 3.0   | 4.0   | 0               |
| 4              | 4.0   | 3.0   | 0               |

Tasks & Questions:

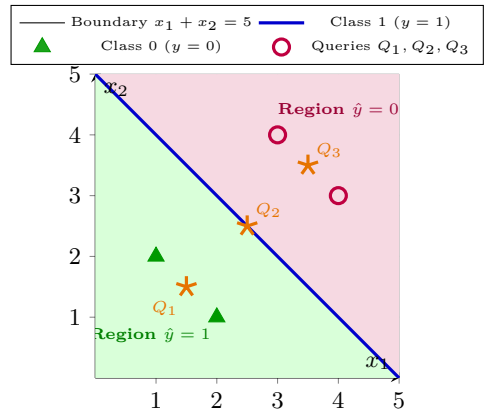
- a) Sketch the decision boundary  $x_1 + x_2 = 5$  on the plot and label regions where  $\hat{y} = 1$  vs  $\hat{y} = 0$ .
- b) Calculate the total Binary Cross-Entropy (BCE) loss under parameter set  $\mathbf{w}^*$ :

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \ln \hat{p}_i + (1 - y_i) \ln(1 - \hat{p}_i)]$$

$\mathcal{L}_{\text{BCE}}^* =$  0.127

- c) Evaluate predictions for test points (Threshold = 0.5; predict  $\hat{y} = 1$  if  $\hat{p} \geq 0.5$ , else 0):
- $Q_1(1.5, 1.5) \implies \hat{p} =$  0.881,  $\hat{y} =$  1
  - $Q_2(2.5, 2.5) \implies \hat{p} =$  0.500,  $\hat{y} =$  1
  - $Q_3(3.5, 3.5) \implies \hat{p} =$  0.119,  $\hat{y} =$  0

Plot points and decision boundary:



Detailed Step-by-Step Solutions & Explanations:

- Decision Boundary and Classification Half-Spaces:** The logit model is  $z(\mathbf{x}) = 5.0 - x_1 - x_2$ . The boundary is formed where the classifier is equally uncertain between both classes:
$$\hat{p} = \sigma(z) = \frac{1}{1 + e^{-z}} = 0.5 \iff z = 0 \iff 5.0 - x_1 - x_2 = 0 \iff \mathbf{x}_1 + \mathbf{x}_2 = 5$$
  - Region  $\hat{y} = 1$  (Below boundary):** For any point with  $x_1 + x_2 < 5$ , the logit  $z > 0$ , so  $\hat{p} > 0.5$ .
  - Region  $\hat{y} = 0$  (Above boundary):** For any point with  $x_1 + x_2 > 5$ , the logit  $z < 0$ , so  $\hat{p} < 0.5$ .
- Binary Cross-Entropy (BCE) Loss Derivation:** For each sample, compute  $z_i$ , predicted probability  $\hat{p}_i = \sigma(z_i) = \frac{1}{1+e^{-z_i}}$ , and sample loss  $\ell_i = -[y_i \ln \hat{p}_i + (1 - y_i) \ln(1 - \hat{p}_i)]$ :

| Sample ( $i$ ) | $(x_{i1}, x_{i2})$ | $y_i$ | $z_i = 5 - x_1 - x_2$ | $\hat{p}_i = \sigma(z_i)$   | Loss Formula       | Loss $\ell_i$ |
|----------------|--------------------|-------|-----------------------|-----------------------------|--------------------|---------------|
| 1              | (1.0, 2.0)         | 1     | +2.0                  | $\sigma(2) \approx 0.8808$  | $-\ln(0.8808)$     | 0.1269        |
| 2              | (2.0, 1.0)         | 1     | +2.0                  | $\sigma(2) \approx 0.8808$  | $-\ln(0.8808)$     | 0.1269        |
| 3              | (3.0, 4.0)         | 0     | -2.0                  | $\sigma(-2) \approx 0.1192$ | $-\ln(1 - 0.1192)$ | 0.1269        |
| 4              | (4.0, 3.0)         | 0     | -2.0                  | $\sigma(-2) \approx 0.1192$ | $-\ln(1 - 0.1192)$ | 0.1269        |

Notice that for all 4 samples, the exact loss is identical:  $-\ln(\sigma(2)) = \ln(1 + e^{-2}) \approx 0.126928$ .

$$\mathcal{L}_{\text{BCE}}^* = \frac{1}{4} \sum_{i=1}^4 \ell_i = \ln(1 + e^{-2}) \approx \mathbf{0.1269} \quad (\approx \mathbf{0.127})$$

- Evaluation and Classification of Test Queries (Threshold = 0.5):**
  - Query  $Q_1(1.5, 1.5)$ :**  $z = 5.0 - 1.5 - 1.5 = +2.0 \implies \hat{p} = \sigma(2.0) \approx \mathbf{0.881} \implies \hat{\mathbf{y}} = \mathbf{1}$  (Class 1).
  - Query  $Q_2(2.5, 2.5)$ :**  $z = 5.0 - 2.5 - 2.5 = 0.0 \implies \hat{p} = \sigma(0.0) = \frac{1}{1+1} = \mathbf{0.500} \implies \hat{\mathbf{y}} = \mathbf{1}$  (since  $\hat{p} \geq 0.5$ ).
  - Query  $Q_3(3.5, 3.5)$ :**  $z = 5.0 - 3.5 - 3.5 = -2.0 \implies \hat{p} = \sigma(-2.0) \approx \mathbf{0.119} \implies \hat{\mathbf{y}} = \mathbf{0}$  (Class 0).

Question 4: *k*-Nearest Neighbors Classification

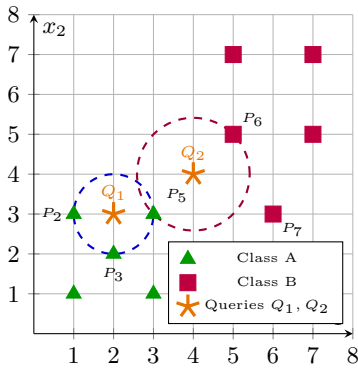
**Distance Metrics & Majority Voting** (compute Euclidean  $L_2$  and Manhattan  $L_1$  distances with majority voting). Below is a dataset of 10 samples belonging to Class A ( $\triangle$ ) and Class B ( $\square$ ), along with two query points  $Q_1(2, 3)$  and  $Q_2(4, 4)$  marked with stars ( $\star$ ). (Note: In case of distance or voting ties, break ties by selecting Class A / lower sample ID.)

Training Samples:

| ID    | Coord  | Class | ID       | Coord  | Class |
|-------|--------|-------|----------|--------|-------|
| $P_1$ | (1, 1) | A     | $P_6$    | (5, 5) | B     |
| $P_2$ | (1, 3) | A     | $P_7$    | (6, 3) | B     |
| $P_3$ | (2, 2) | A     | $P_8$    | (7, 5) | B     |
| $P_4$ | (3, 1) | A     | $P_9$    | (5, 7) | B     |
| $P_5$ | (3, 3) | A     | $P_{10}$ | (7, 7) | B     |

Tasks & Questions:

- a) For  $Q_1(2, 3)$ , find the 3 nearest neighbors using **Euclidean distance**  $d_E = \sqrt{\Delta x_1^2 + \Delta x_2^2}$ :  
Neighbors: {  $P_2, P_3, P_5$  }  $\implies \hat{y}(Q_1) =$  A
- b) For  $Q_2(4, 4)$ , classify using **Euclidean distance**:  
  - $k = 1$ : Neighbor =  $P_5$   $\implies \hat{y} =$  A
  - $k = 3$ : Neighbors =  $P_5, P_6, P_7$   $\implies \hat{y} =$  B
- c) For  $Q_2(4, 4)$ , classify using **Manhattan distance**  $d_M = |\Delta x_1| + |\Delta x_2|$  with  $k = 3$ :  
Neighbors: {  $P_5, P_6, P_7$  }  $\implies \hat{y}(Q_2) =$  B



Detailed Step-by-Step Solutions & Explanations:

• Comprehensive Distance Calculation Table:

| ID       | Class | Coord  | $d_E(P_i, Q_1(2, 3))$               | $d_E(P_i, Q_2(4, 4))$               | $d_M(P_i, Q_2(4, 4))$            |
|----------|-------|--------|-------------------------------------|-------------------------------------|----------------------------------|
| $P_1$    | A     | (1, 1) | $\sqrt{1^2 + 2^2} = 2.236$          | $\sqrt{3^2 + 3^2} = 4.243$          | $ 1 - 4  +  1 - 4  = 6$          |
| $P_2$    | A     | (1, 3) | $\sqrt{1^2 + 0^2} = \mathbf{1.000}$ | $\sqrt{3^2 + 1^2} = 3.162$          | $ 1 - 4  +  3 - 4  = 4$          |
| $P_3$    | A     | (2, 2) | $\sqrt{0^2 + 1^2} = \mathbf{1.000}$ | $\sqrt{2^2 + 2^2} = 2.828$          | $ 2 - 4  +  2 - 4  = 4$          |
| $P_4$    | A     | (3, 1) | $\sqrt{1^2 + 2^2} = 2.236$          | $\sqrt{1^2 + 3^2} = 3.162$          | $ 3 - 4  +  1 - 4  = 4$          |
| $P_5$    | A     | (3, 3) | $\sqrt{1^2 + 0^2} = \mathbf{1.000}$ | $\sqrt{1^2 + 1^2} = \mathbf{1.414}$ | $ 3 - 4  +  3 - 4  = \mathbf{2}$ |
| $P_6$    | B     | (5, 5) | $\sqrt{3^2 + 2^2} = 3.606$          | $\sqrt{1^2 + 1^2} = \mathbf{1.414}$ | $ 5 - 4  +  5 - 4  = \mathbf{2}$ |
| $P_7$    | B     | (6, 3) | $\sqrt{4^2 + 0^2} = 4.000$          | $\sqrt{2^2 + 1^2} = \mathbf{2.236}$ | $ 6 - 4  +  3 - 4  = \mathbf{3}$ |
| $P_8$    | B     | (7, 5) | $\sqrt{5^2 + 2^2} = 5.385$          | $\sqrt{3^2 + 1^2} = 3.162$          | $ 7 - 4  +  5 - 4  = 4$          |
| $P_9$    | B     | (5, 7) | $\sqrt{3^2 + 4^2} = 5.000$          | $\sqrt{1^2 + 3^2} = 3.162$          | $ 5 - 4  +  7 - 4  = 4$          |
| $P_{10}$ | B     | (7, 7) | $\sqrt{5^2 + 4^2} = 6.403$          | $\sqrt{3^2 + 3^2} = 4.243$          | $ 7 - 4  +  7 - 4  = 6$          |

- **Part (a):  $Q_1(2, 3)$  with Euclidean Distance ( $k = 3$ ):**
  - The three smallest Euclidean distances to  $Q_1(2, 3)$  are  $d(P_2) = 1.000$ ,  $d(P_3) = 1.000$ , and  $d(P_5) = 1.000$ . (Next nearest is  $P_1$  and  $P_4$  with  $d = \sqrt{5} \approx 2.236$ ).
  - 3 Nearest Neighbors:  $\{P_2, P_3, P_5\}$ .
  - Voting:  $P_2 \in \text{Class A}$ ,  $P_3 \in \text{Class A}$ ,  $P_5 \in \text{Class A} \implies \mathbf{3}$  votes for Class A,  $\mathbf{0}$  votes for Class B.
  - Classification:  $\hat{\mathbf{y}}(\mathbf{Q_1}) = \text{Class A (or A)}$ .
- **Part (b):  $Q_2(4, 4)$  with Euclidean Distance ( $k = 1$  and  $k = 3$ ):**
  - **Ranked Distances to  $Q_2(4, 4)$ :** 1st & 2nd:  $P_5$  ( $\sqrt{2} \approx 1.414$ , Class A) and  $P_6$  ( $\sqrt{2} \approx 1.414$ , Class B) [Tie!]; 3rd:  $P_7$  ( $\sqrt{5} \approx 2.236$ , Class B); 4th:  $P_3$  ( $\sqrt{8} \approx 2.828$ , Class A).
  - **Case  $k = 1$  (Tie-Breaking):** An exact distance tie occurs between  $P_5$  and  $P_6$  at distance 1.414. By the tie-breaking rule (“select Class A / lower sample ID”),  $P_5$  (Class A, sample ID  $5 < 6$ ) is chosen  $\implies$  Neighbor is  $\mathbf{P_5} \implies \hat{\mathbf{y}} = \text{Class A}$ .
  - **Case  $k = 3$  (Majority Voting):** The 3 nearest neighbors are  $\{P_5, P_6, P_7\}$ . Class counts:  $P_5 \in \text{Class A}$  (1 vote),  $P_6 \in \text{Class B}$  (1 vote),  $P_7 \in \text{Class B}$  (1 vote). Total votes:  $\mathbf{1}$  vote Class A,  $\mathbf{2}$  votes Class B  $\implies$  Majority is Class B  $\implies \hat{\mathbf{y}} = \text{Class B}$ .
- **Part (c):  $Q_2(4, 4)$  with Manhattan Distance ( $k = 3$ ):**
  - Using  $d_M = |\Delta x_1| + |\Delta x_2|$ , the sorted distances are:  $d_M(P_5) = 2$ ,  $d_M(P_6) = 2$ ,  $d_M(P_7) = 3$ . The next nearest points ( $P_2, P_3, P_4, P_8, P_9$ ) all have  $d_M = 4$ .
  - The 3 nearest neighbors are  $\{P_5, P_6, P_7\}$ .
  - Majority vote:  $P_5 \in \text{Class A}$  (1 vote),  $P_6 \in \text{Class B}$  (1 vote),  $P_7 \in \text{Class B}$  (1 vote)  $\implies \mathbf{1}$  vote Class A,  $\mathbf{2}$  votes Class B  $\implies \hat{\mathbf{y}}(\mathbf{Q_2}) = \text{Class B}$ .